P-192 label-descent extraction and invariant construction target

Cheap label invariant
Check whether it descends to the represented point

Nonconstant polynomial descent, degree below r
Extract x from two leading coefficients; verify xG=Q

Missing construction interface
Produce the polynomial from G,Q without already knowing x

Derived reduction. Let G have prime order r and Q=xG. Let H(A,B) be an explicitly represented nonconstant polynomial over F_r of total degree d<r. Suppose H(a,b)=H(c,e) whenever aG+bQ=cG+eQ, for every scalar pair. Then H(A,B)=h(A+xB) as a polynomial identity. Its coefficients yield x by one field division after scanning the representation. This is a reduction with an invariant-construction premise, not an algorithm for constructing that invariant. [pcc:unverified]

Proof of descent. Set U=A+xB and V=B, and F(U,V)=H(U-xV,V). All representations (u-xv,v) correspond to uG, hence F(u,v)=H(u,0) for all u,v in F_r. The difference has total degree at most d<r. A polynomial of degree below r in each variable that vanishes on the full r-by-r grid is the zero polynomial, by applying the univariate root bound twice. Therefore F(U,V)=h(U), where h(U)=H(U,0), and H(A,B)=h(A+xB). Since H is nonconstant, h has some degree d>=1 and leading coefficient c!=0. [pcc:unverified]

Explicit extractor. Write C_d for the coefficient of A^d and C_{d-1,1} for that of A^(d-1)B. Expansion gives C_d=c and C_{d-1,1}=cdx. Since d<r, d is invertible, so x=C_{d-1,1}/(d C_d) modulo r. For an arbitrary supplied polynomial, verify the whole identity H_B=x H_A and finally check [x]G=Q. Under total degree below r, the derivative identity also implies descent: after the same substitution F_V=0, so F is independent of V. Dense input scanning costs O(d^2) coefficient operations; sparse scanning is linear in the supplied term count. Coefficient size, representation and construction costs are not free. [pcc:unverified]

Order-five condition. Suppose additionally H(lambda A,lambda B)=H(A,B) for the nontrivial fifth root lambda used in the P-192 construction. The same full-grid root bound makes this a polynomial identity. Since h(lambda U)=h(U), every nonzero coefficient of h has degree divisible by five. Thus a nonconstant descending invariant has degree at least five. At degree five it is c0+c5(A+xB)^5, with x recovered from the A^4B and A^5 coefficients. The six homogeneous coefficients lie on a binomially scaled degree-five Veronese curve; guessing an arbitrary homogeneous invariant does not enforce the needed descent. [pcc:unverified]

Counterexample and controls. H=A^5 is invariant under simultaneous multiplication of both labels by lambda. For the public scalar x=17, labels (1,0) and (-16,1) both represent G, but their H values differ. Consequently order-five invariance by itself cannot define a partition of elliptic-curve points. The checker constructs 20 public descending invariants with degrees 5, 10, 15 and mixed support, for x in {0,1,2,17,20261107}. A separate checker extracts x from the top-degree coefficients and expands H(U-xV,V), checking cancellation of every positive V coefficient. Constant, B^5, and A^5+B^5 inputs exercise rejection paths. These are exact fixtures, with no unknown scalar in their construction. [pcc:unverified]

Separate coordinate-degree bound. For P-192, r<p. Suppose f(T) in F_p[T] interpolates x([k]G) for every integer 1<=k<r. Write G=(u,v) and E:y^2=x^3+a_E x+b_E. For k=2,...,r-2 the addition law gives (f(k+1)+f(k-1))(f(k)-u)^2=2[u f(k)^2+(a_E+u^2)f(k)+a_E u+2b_E]. There are r-3 distinct arguments in F_p, and no denominator vanishes there. If f has positive degree d, the difference polynomial has degree exactly 3d: its leading coefficient is twice the cube of the leading coefficient of f, whereas the right side has degree at most 2d. It cannot vanish at r-3 points unless 3d>=r-3. A constant f cannot represent all nonzero multiples of G, since a fixed x-coordinate has at most two curve points. Therefore d>=ceil((r-3)/3), which is 20923672451288935879452631410586863379223982577276140946 93 for P-192. The numerator identity is checked symbolically. [pcc:unverified]

Field boundary. The label-descent reduction is over F_r. The coordinate interpolation bound is over F_p with integer indices before wraparound. There is no field embedding identifying these two fields. The latter bound does not rule out an arbitrary F_r encoding, an oracle, or a succinct high-degree function; it only excludes the specified low-degree F_p coordinate interpolation proposal. The written proofs are algebraic derivations, not machine-checked universal theorems. [pcc:unverified]

Next construction. Search for a point-computable function with a succinct scalar-field polynomial representation and certified descent, or a different equivariant transition avoiding such representation. A candidate must supply its

construction from G,Q, its evaluation cost, and its coefficient/extractor interface. The smaller class number or L-value does not by itself supply any of these interfaces. No novelty claim is made for these elementary reductions.

Literature context. Galbraith, Mathematics of Public Key Cryptography, section 14.4, discusses equivalence-class walks: https://www.math.auckland.ac.nz/~sgal018/crypto-book/main.pdf . That context motivates descent to a point orbit; the explicit polynomial reduction and field distinction above are proved here from their stated hypotheses. [pcc:unverified]